

ML Foundations — Final Project Report

Adult / Census Income — Targeted Marketing Model | Qasim | random_state = 42

1. Problem Definition

A marketing team wants to target a premium offer at households likely to earn over \$50K/year, instead of contacting everyone — controlling outreach cost while still reaching the intended audience. The business objective is to maximize campaign ROI by minimizing wasted contacts (false positives) while still catching genuine high-income prospects (avoiding false negatives), using a model that ranks households by likelihood of high income.

2. Data Preparation

Source: UCI Adult / Census Income (1994 U.S. Census), 48,842 rows, 14 raw features, via `sklearn.datasets.fetch_openml('adult', version=2)`. **Target:** income >50K = 1, ≤50K = 0 (23.93% positive base rate). **Preprocessing:** median imputation + StandardScaler for numeric features; most-frequent imputation + OneHotEncoder(handle_unknown='ignore') for categorical features, all inside a single ColumnTransformer/Pipeline fit only on training data. **Feature engineering:** 8 row-wise features (age_bucket, hours_bucket, capital_gain_flag, capital_gain_log, higher_education, education_x_hours, age_x_hours, marital_married), each justified by a plausible income-driving mechanism and verified with mutual-information/grouped-mean signal checks; capital_gain_flag was later dropped due to collinearity with capital_gain_log. **Validation strategy:** stratified 70/10/20 train/dev/test split (random_state=42), fixed from Day 1 and reused identically through Day 5, plus StratifiedKfold(5) for all cross-validated comparisons.

3. Model Development

Models tested across the project: Logistic Regression, Decision Tree, Random Forest, and HistGradientBoosting. Day 1 baselines: majority-class (ROC AUC 0.500) and a rule (education-num≥13: ROC AUC 0.719). Day 3 5-fold CV with the full engineered feature set:

Model	CV ROC AUC	CV F1
Logistic Regression	0.9114 ± 0.0032	0.6679 ± 0.0047
Random Forest	0.9006 ± 0.0027	0.6628 ± 0.0051
HistGradientBoosting	0.9262 ± 0.0036	0.7031 ± 0.0082

HistGradientBoosting led on every metric; a paired t-test vs. Logistic Regression gave $p=0.0001$ (every fold favored it), so the difference was treated as real, not noise.

4. Model Tuning

RandomizedSearchCV with StratifiedKfold, scoring='roc_auc'. **Compute constraint:** this project ran on a single CPU core; a benchmark found Logistic Regression's saga solver with L1 penalty took ~80s per fit vs. ~0.3s for liblinear, so saga was dropped (liblinear covers both L1/L2) and n_iter/cv were reduced from the initial 50–100/5 plan to keep the search runnable (documented, not silently narrowed in parameter range).

Model	Search space	n_iter × cv	Best CV ROC AUC	Best parameters
Logistic Regression	penalty(l1/l2), C (log-uniform 1e-3–1e2), solver=liblinear, max_iter	30 × 3	0.9111	penalty=l1, C=0.541, max_iter=2000
HistGradientBoosting	learning_rate, max_iter, max_depth, l2_regularization, min_samples_leaf	15 × 3	0.9253	lr=0.028, max_iter=200, max_depth=None, l2=0.0054, min_samples_leaf=20

5. Diagnostics

Learning curve (HistGradientBoosting, tuned): train ROC AUC 0.9405 vs. validation 0.9253 (gap 0.015) — small and stable, no meaningful overfitting or underfitting. **Validation curves:** Logistic Regression's train/validation gap grows with C, peaking validation score at C=1.0; HistGradientBoosting's gap shrinks monotonically as min_samples_leaf increases (0.009 at 20 down to 0.006 at 400), confirming it as the expected regularization lever. **Calibration:** Brier score 0.0865 before, 0.0862 after CalibratedClassifierCV(sigmoid, cv=5) — kept, since it improved (the improve-or-skip rule).

6. Final Results

Stage	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Day 1 Baseline B	0.7530	0.4844	0.4970	0.4906	0.7189	0.4443
Day 2 Logistic Regression	0.8527	0.7431	0.5877	0.6563	0.9039	0.7624
Day 4 tuned, test, thr=0.50	0.8760	0.7930	0.6523	0.7158	0.9279	0.8302

Day 5 final, test, thr=0.40	0.8692	0.7253	0.7297	0.7275	0.9279	0.8302
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Test confusion matrix at threshold 0.40: TN=6785, FP=646, FN=632, TP=1706. Test ROC AUC (0.9279) matches the tuned CV estimate (0.9253) closely and even exceeds Day 3's untuned CV estimate (0.9262) — performance is stable from CV to held-out test, with no sign the search overfit to CV folds. The Day 5 reload of the saved artifact reproduced Day 4's test numbers exactly, confirming full determinism.

7. Model Interpretation

Permutation importance (ROC AUC drop) on the deployed pipeline: **marital-status** (0.1024) dominates, followed by **capital-gain** (0.0604), **age** (0.0414), and **education-num** (0.0352) — consistent with Logistic Regression coefficients and Random Forest importances from earlier days, a cross-method sanity check. **Gender bias finding:** recall is 0.670 for Female vs. 0.741 for Male on the test set, yet `sex` itself ranks only 10th of 14 in direct permutation importance (0.0010) — the disparity is **mediated through correlated features** (marital-status, occupation, hours-per-week differ by gender in this data), not the model reading sex directly. This means removing the sex column alone would likely not fix the gap. **Non-monotonic capital-gain finding:** a sensitivity sweep found specific dollar amounts (7298, 15024, 99999) heavily overrepresented in the >50K class in the original extract, while others (2174, 3325, 4650, 5013) are overrepresented in ≤50K — the model learned these training-data-specific values rather than a smooth relationship, so arbitrary capital-gain values (e.g., exactly \$5,000) can produce unstable predictions even with an otherwise strong profile.

8. Production Readiness

Artifact: final_pipeline.pkl (~4.5MB), a CalibratedClassifierCV wrapping the full tuned HistGradientBoosting pipeline (feature engineering → preprocessing → model) — accepts a raw DataFrame directly. **Inference:** load with joblib.load(), call predict_proba(), apply threshold 0.40 — no separate preprocessing step needed or permitted (the pipeline owns it).

Reproducibility: random_state=42 throughout; exact library versions pinned in requirements.txt (scikit-learn 1.8.0, pandas 3.0.2, numpy 2.4.4, scipy 1.17.1, joblib 1.5.3); the identical stratified split is regenerated deterministically in every day's notebook.

9. Limitations & Future Improvements

Data: more granular occupation/industry codes and a tenure/experience feature would likely help — false negatives skew toward workers without a formal-education signal, suggesting experience-driven income isn't well captured. **Models:** XGBoost/LightGBM or a small neural network could be benchmarked against HistGradientBoosting; an ensemble of Logistic Regression + HistGradientBoosting is worth testing since their errors aren't perfectly correlated. **Validation:** this is 1994 cross-sectional census data with no time dimension, so time-based validation isn't applicable here, but any future refresh with newer/longitudinal data should validate temporally rather than reuse a single random split. **Fairness/ethics:** the gender recall gap documented in Section 7 is a real, mediated-through-correlated-features disparity that should block real deployment until addressed via a fairness-aware technique (reweighting, constrained optimization, or subgroup-specific thresholds) rather than simple feature removal; the capital-gain non-monotonicity should also be addressed (e.g., by re-bucketing the feature) before trusting the model on data outside this training distribution.